

Sonia Irakoze

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EDUCATION

University of Rochester, Hajim School of Engineering & Applied Sciences

Rochester, NY

Bachelor of Science in Mechanical Engineering

Expected May 2028

- Alan & Jane Handler Endowed Scholar (top ~1% of incoming class) • Kearns Scholar • Dean's List, Fall 2024
- Coursework: Thermodynamics, Engineering Mechanics, Mechanical Fabrication, CAD & Drawing, Electricity & Magnetism

EXPERIENCE

Colorado Department of Transportation (CDOT), Staff Bridge Branch

Denver, CO

Engineering Intern, Bridge Asset Management

Summer 2026

- Wrote SQL queries (SSMS) on inspection, project, and clearance databases supporting Colorado's 3,500+ bridge inventory
- Reviewed structural plan sets to classify bridge treatment types and update records to National Bridge Inventory standards
- Cross-checked structure IDs across SIMSA, OnBase, and OTIS to fix discrepancies; joined field bridge inspections
- Built **CDOT Compass**, a React prototype mapping CDOT divisions for new staff; published as a 2026 CDOT Lean idea

AGR Sensors

Rochester, NY (Remote)

Data Research Intern, CropSight AI Pipeline

January 2026 – Present

- Audited 250+ source chunks from 22 agricultural reference books for CropSight, an AI crop-disease diagnosis tool
- Catalogued 212 disease and pest entries as ML-ready JSON, with Python checks reaching zero schema errors

Ugandan Water Project

Lima, NY

Development Intern, Schwartz Community-Based Internship Scholar

June 2025 – August 2025

- Contacted 100+ donors and corporate sponsors, helping raise \$5K+ for clean-water projects in Uganda
- Cleaned donor-database records and ran communications for the organization's annual golf fundraiser

University of Rochester River Campus Libraries

Rochester, NY

Editorial Assistant Intern

May 2025 – August 2025

- Managed metadata, DOIs, and stylesheets for three academic journals; designed the Summer 2025 *Blake Quarterly* cover

University of Rochester IT Center

Rochester, NY

Level 2 Training Coordinator (promoted from Help Desk Technician)

January 2025 – Present

- Resolve escalated network, authentication, and malware issues (75+ tickets/week); mentor and train Level 1 staff

PROJECTS

SAquaSolve Rwanda

Gashora, Rwanda

Founder & Legal Representative

August 2023 – Present

- Won a \$10,000 Davis Projects for Peace grant (2026) to develop a solar-powered system treating water from Lake Mirayi
- Led two-day community water and hygiene workshops; coordinating lake water-quality testing with a research partner
- Developed a conceptual layout with an engineer: solar pumping, containerized treatment, storage, and community access points

University of Rochester CubeSat & Yellowjackets Aero Design

Rochester, NY

Structures Team Member (CubeSat); Design Lead (Aero Design)

April 2025 – Present

- Model CubeSat frame parts in Siemens NX, balancing mass against launch loads; resolve fit issues with avionics and propulsion
- Compared glider wing and tail designs with FEA and built laser-cut prototypes over 15+ iterations; team cut weight ~10%

LEADERSHIP

NSBE Region 1 Upstate Zone Publications Chair (2026–27) and UR Chapter Senator (2025–26) • Peer Advisor • Student Alumni Ambassador • International Student Mentor • Co-founder, UniBooth (iZone grant) • TechGirls Alumnae Advisory Council

AWARDS

Greene Center Internship Funding (2026) • U.S. State Department TechGirls Program (2023) • Silver Award, Queen's Commonwealth Essay Competition (2023) • 3rd Prize, East African Community Essay Competition (2022)

SKILLS

Engineering: Siemens NX, Fusion 360, FEA, GD&T, OpenRoads Designer; lathe, mill, sheet metal, 3D printing, laser cutting

Software & Design: SQL, Python, MATLAB, Git, HTML/CSS, React/TypeScript; Adobe InDesign, Illustrator, Figma